

ARYENDRA PRATAP SINGH

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Summary

Computer Science undergraduate who builds and ships AI, full-stack, and data systems, from a deployed LLM document reviewer to RAG search and SQL analytics, backed by two information-systems internships. Seeking software, AI/ML, data, and IT internships and co-ops.

Education

Lakehead University – Honours B.Sc. in Computer Science Thunder Bay, ON | Expected April 2027
CGPA: 78.31% | **Coursework:** Data Structures, Algorithms, Operating Systems, DBMS, Software Engineering

Technical Skills

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, HTML, CSS

AI and Data: LLM APIs (Gemini, OpenAI), RAG, LangChain, prompt engineering, OCR, spaCy, NLTK

Backend and Databases: FastAPI, REST APIs, Pydantic, SQLAlchemy, JWT, MySQL, PostgreSQL, Redis

Frontend and Cloud: React, Next.js, Tailwind CSS, Vite, Docker, Render, Vercel, Git, GitHub, Linux

IT and Process: Help desk ticketing, troubleshooting, technical documentation, SDLC, Microsoft 365, Excel

Technical Projects

RegenMed Form Checker | *Python, FastAPI, Gemini API, Pydantic, NumPy, React, TypeScript, Render*

- Built and deployed an AI document reviewer at the Thunder Bay AI Hackathon (Sept 2026), flagging missing signatures, initials, and dates on a tissue bank's scanned regulatory forms.
- Engineered multimodal Gemini extraction into Pydantic JSON Schemas feeding a 23-rule Python engine that traces every error to its exact page, section, row, and field.
- Reduced review time by ~75% (85 s to 21 s) through model benchmarking, concurrent API calls, and a NumPy layout classifier that scored 34/34 correct matches without an API call.
- Added parallel "second opinion" reads that route AI disagreements to a human-review list.
- Generated staff-ready correction notes with a second LLM API and a two-provider fallback for outages.
- Secured API keys in environment variables and built a CLI test harness logging latency, retries, and errors.
- Deployed React and FastAPI as one Render cloud service with health checks and 400/413/503 error handling.

AI Real Estate Assistant | *Python, FastAPI, Next.js, React, LangChain, ChromaDB, PostgreSQL, Redis, Docker*

- Built a full-stack AI platform for conversational property search, market analytics, and mortgage calculators.
- Engineered a RAG pipeline with LangChain, ChromaDB, and FastEmbed for context-aware property search.
- Unified 6 LLM providers (OpenAI, Gemini, Claude, DeepSeek, Grok, Ollama) behind one conversational framework.
- Built a responsive Next.js chat interface with real-time backend communication and saved user preferences.
- Developed modular FastAPI services with JWT auth, sessions, and SQLAlchemy/Alembic schema migrations.

DocumentAI – Intelligent Document Processing | *Python, Tesseract OCR, spaCy, NLTK, SQL*

- Developed a Tesseract OCR pipeline converting scanned PDFs and images into machine-readable text for analysis.
- Designed spaCy and NLTK pipelines for entity recognition, keyword extraction, and classification into SQL storage.
- Added AI summarization and semantic search, improving document review and information retrieval.

HELOC Portfolio Analytics Database | *MySQL, SQL, Stored Procedures, Triggers, Views*

- Designed a normalized MySQL schema for customers, accounts, and payments, enforcing integrity with constraints.
- Built stored procedures, triggers, and views over 150+ records to automate delinquency and roll-rate reporting.

IT Help Desk Simulation and Ticketing Workflow | *Excel, Technical Documentation*

- Designed a ticketing workflow for intake, prioritization, escalation, and resolution with written procedures.
- Analyzed ticket trends in Excel to identify recurring issues and recommend support process improvements.

Professional Experience

Stock Unloader Associate | **Walmart Canada**, Thunder Bay, ON May 2026 – Present

- Process high-volume overnight inventory to team targets, maintaining cold-chain compliance and safety inspections.
- Label and manage overstock inventory and assist customers as part of a high-volume overnight team.

Part-Time Team Member | **Masala Grille**, Thunder Bay, ON Oct 2024 – Apr 2025

- Delivered fast-paced customer service, prioritizing shifting requests while maintaining service quality.

Intern | **MyGov**, Delhi, India May 2022 – Jul 2022

- Digitized and validated 600+ public records with consistent naming and metadata for reliable retrieval.
- Generated structured reports and datasets, identifying outdated and duplicate records to support decisions.
- Contributed to process improvements that reduced manual effort by ~20% while improving documentation accuracy.

Intern | **Eisenvault**, Delhi, India May 2021 – Jul 2021

- Validated records and documented workflows in enterprise content systems, resolving inconsistencies.